

Matt Bowring

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[Website](#) | [GitHub](#) | [LinkedIn](#)

Experience

MathWorks

Natick, MA (on-site)

Software Engineer, Quantum Computing

May 2022 - Present

- Lead development of the [MATLAB Quantum Computing Library](#) (11k+ users, Windows/Linux/macOS support)
 - Designed core object-oriented MATLAB framework for building, simulating, and executing parameterized quantum circuits
 - Developed templated C++ kernels for multi-threaded state-vector simulation on CPU/GPU, delivering 30x speedup over MATLAB implementation
 - Integrated C++ graph compiler for qubit placement, routing, and hardware-specific assembly code generation with batched execution support
 - Built client-side REST interfaces to AWS and IBM quantum cloud services with authentication and job management to enable quantum hardware access directly from MATLAB
 - Implemented numerical methods to estimate sparse Pauli observables, synthesize quantum logic gates, and mitigate measurement errors
 - Owned 50k+ line codebase from prototype to production, including CI/CD, test infrastructure, and roadmap planning for monthly releases
- Collaborated with industry partners on quantum computing applications, with results presented at SPIE, ACS, and AIChE conferences
 - Developed quantum-classical machine learning workflows using methods such as QAOA/VQE, quantum Monte Carlo, Pauli correlation encoding, and error correction codes

Senior Associate Software Engineer

May 2021 - May 2022

- Implemented machine learning and quantum computing experiments for customer research applications
 - Trained temporal graph neural network on 1M+ molecular instances to predict energy of equilibrium structures
 - Implemented Hamiltonian simulations of superconducting qubits under transverse and longitudinal coupling fields
 - Formulated Hamiltonians with penalty constraints for coarse-grained protein folding and benchmarked performance on quantum annealing hardware
- Supported 100+ customers using MATLAB products, including code design, debugging, and performance improvements

Education

Purdue University

West Lafayette, IN (remote)

M.S. Electrical and Computer Engineering (3.8 GPA)

Expected May 2027

- Designed and fabricated a mixed-signal oscillator-based Ising machine for combinatorial optimization research ([arXiv](#))
 - Developed circuit simulations in LTspice and MATLAB, designed schematics and PCB layouts with Altium Designer, and verified hardware components with oscilloscope and waveform generator
 - Developed embedded C++ firmware using SPI and FreeRTOS to enable programmable oscillator couplings and real-time phase measurement
- Applied random matrix theory and nonequilibrium thermodynamics to analyze annealing schedules for oscillator-based Ising machines

The University of New Hampshire

Durham, NH (on-site)

B.S. Mechanical Engineering (3.9 GPA)

May 2021

- Awarded \$100,000+ scholarships, led Quadcopter Engineering Team, and volunteered as Makerspace administrator

Projects

- Optimized matrix multiplication kernels in C using ARM NEON intrinsics, cache tiling, and register blocking
- Developed socket-based networking protocols in Python with selective retransmission and DNS resolution
- Built ROS/MAVROS interface to quadcopter flight controller for autonomous waypoint tracking, validated with Gazebo software-in-the-loop simulation
- Trained recurrent convolutional neural network on dash-camera driving footage to predict frame misalignment with 86% accuracy

Skills

Languages: MATLAB, C/C++, Python, Shell Scripting

Libraries: PyTorch, PennyLane, Qiskit, NetworkX

Tools: CMake, OpenMP, LLVM/MLIR, CUDA, Git, GoogleTest, Docker, Nix